Yasas Senarath Wijesuriya

Robust AI & Machine Learning | LLM Evaluation | Research Software Systems

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Research Interests

Robust and adaptive machine learning, continual learning, knowledge-guided AI, LLM evaluation, retrieval-augmented generation, and research software systems, with an emphasis on developing reliable AI methods for dynamic, real-world environments.

Education

PhD in Information Technology May 2026
George Mason University, Fairfax, VA
Thesis: *Modeling Evolving Human Behavior in Online Data Streams via Knowledge-Guided Continual Learning*
Advisor: Dr. Hemant Purohit

Master of Science in Applied Information Technology Aug 2022
George Mason University, Fairfax, VA
Concentration: Cyber-Human Systems (CBHS)
Advisor: Dr. Hemant Purohit

Bachelor of Science in Engineering (Honours) Jan 2018
University of Moratuwa, Sri Lanka
Advisor: Dr. Surangika Ranathunga GPA: 3.95/4.2

Research Experience

Graduate Research Assistant, Humanitarian Informatics Lab, George Mason University Aug 2019 – Aug 2025

- Developed KGCL, a knowledge-guided continual learning framework for detecting evolving behaviors in social media streams; improved AUC by 2.13 percentage points and reduced forgetting by 2.5 points versus baselines
- Supervised development of a contrastive-learning validation framework for LLM-generated annotations; improved verification accuracy by 3.7 percentage points over the baseline on a deviant-behavior classification task
- Directed a GraphRAG natural-language querying pipeline from prototype to structured knowledge-graph retrieval, mentoring one student researcher through development
- Engineered SOCYTI, a real-time pipeline and dashboard co-designed with NGO and government stakeholders; processed 26,000+ tweets using language models and network centrality methods
- Deployed CitizenHelper-CERT, a real-time COVID-19 risk analytics system that scores social media posts using trained ML models (0.81 F1), surfaced to emergency managers via a Kibana dashboard with semantic search
- Built CROME, a practitioner-first crowdsourced incident detection system that outperformed a Bayesian fusion baseline by 17–29 percentage points in F1 across spatial scales (IEEE ICDM 2021; Digital Government 2024)
- Led ETL and exploratory analysis across six spatiotemporal police and fire data sources for a Transport Risk Early Warning System developed with the City of Virginia Beach

Summer Research Assistant, Center for Climate Change Communication, George Mason University Jun 2021 – Aug 2021

- Designed EvoLyzer to analyze 490,000+ tweets from 738 participants and measure climate-change education impact through changes in communication behavior (NOCAPS 2024)

Research Assistant, DataSEARCH, University of Moratuwa Jan 2018 – Jul 2019

- Developed a multi-embedding deep learning model for implicit emotion classification of tweets; ranked 8th of 30 teams in the WASSA-2018 shared task, outperforming the baseline by 8pp (68.0% macro-F1)
- Built transfer learning models for emotion intensity prediction (SemEval-2018 Task 1); outperformed state of the art for 2 of 4 emotions (avg. Pearson 0.798) with saliency-map-based explainability

Professional Experience

Graduate Teaching Assistant, Information Sciences and Technology, George Mason University Aug 2025 – May 2026

- Conducted hands-on sessions and evaluated coursework for 70+ students across Database Fundamentals, Data Analytics in Social Media, and Introduction to Research in Applied IT

Visiting Instructor, Computer Science and Engineering, University of Moratuwa Jan 2018 – Jul 2019

- Co-taught a presentation skills course and led data science labs for 50+ undergraduate Computer Science and Engineering students

Trainee Software Engineer, CodeGen International (PVT) Ltd. Jul 2016 – Dec 2016

- Built an NLP pipeline (GATE, Apache Spark) for opinion and named entity extraction from tourist attraction reviews as an R&D project for TravelBox, a global travel platform product

Technical Expertise

Research Methods: Experimental Design, Statistical Inference, A/B Testing, Model Evaluation, Reproducible Research, Research Proposal Development

ML & NLP: PyTorch, PyTorch Lightning, Transformers (Hugging Face), scikit-learn, spaCy, Gensim, NLTK

AI/ML Techniques: Continual Learning, Contrastive Learning, Transfer Learning, RAG, LLM Fine-Tuning, Agentic Systems, Prompt Engineering, Network Analysis

Search & Retrieval: FAISS, ChromaDB, Elasticsearch (Semantic Search)

Programming: Python, SQL, HTML, CSS, JavaScript, Java

Data & Systems: Pandas, NumPy, GeoPandas, MLflow, Matplotlib, Seaborn, Plotly, Dash, Tableau, Kibana, Gephi, FastAPI, Flask, Docker, Git, CI/CD, SLURM

Manuscripts Under Review

[1] **Yasas Senarath** and Hemant Purohit. "KGCL: Knowledge-guided Continual Learning for Detecting Evolving Human Behaviors in Online Communities". In: (2026). Under Review.

[2] **Yasas Senarath** and Hemant Purohit. "Evaluating Class-Incremental Learning Methods for Robust Subjective Text Analytics Systems". In: (2025). Under Review.

Peer-Reviewed Publications

Continual Learning

[3] **Yasas Senarath** and Hemant Purohit. "Knowledge-guided Continual Learning for Behavioral Analytics Systems". In: *The 7th IEEE International Conference on Cognitive Machine Intelligence* (*CogMI*). 2025.

[4] **Yasas Senarath** and Hemant Purohit. "Lifelong Learning Framework for Multilingual Hate Speech Detection in Social Media Streams". In: *Proceedings of the 17th International Conference on Social Computing, Behavioral-Cultural Modeling, & Prediction and Behavior Representation in Modeling and Simulation* (*SBP-BRiMS*). 2024.

Emergency Incident Detection

[5] **Yasas Senarath**, Ayan Mukhopadhyay, Hemant Purohit, and Abhishek Dubey. "Designing a Human-Centered AI Tool for Proactive Incident Detection Using Crowdsourced Data Sources to Support Emergency Response". In: *Digital Government: Research and Practice* 5.1 (2024), pp. 1–19.

[6] **Yasas Senarath** et al. "Practitioner-Centric Approach for Early Incident Detection Using Crowdsourced Data for Emergency Services". In: *2021 IEEE International Conference on Data Mining* (*ICDM*). IEEE. 2021, pp. 1318–1323.

[7] **Yasas Senarath**, Saideep Nannapaneni, Hemant Purohit, and Abhishek Dubey. "Emergency Incident Detection from Crowdsourced Waze Data Using Bayesian Information Fusion". In: *2020 IEEE/WIC/ACM International Joint Conference on Web Intelligence and Intelligent Agent Technology* (*WI-IAT*). IEEE. 2020, pp. 187–194.

Streaming Data Modeling

[8] **Yasas Senarath**, Amanda Borth, Edward Maibach, and Hemant Purohit. "EVO-LYZER: Social Media Mining System for Evolving Communication Behavior Analytics to Aid Climate Change Programs". In: *Workshop: NOCAPS 2024 - Networks and Opinions on Climate Action in the Public Sphere* (*NOCAPS*). 2024.

[9] **Yasas Senarath**, Hemant Purohit, and Rajendra Akerkar. "Violence-Inducing Behavior Prevention in Social-Cyber Space". In: *International Symposium on AI, Data and Digitalization*. Springer Nature Switzerland Cham. 2023, pp. 151–159.

[10] **Yasas Senarath**, Antonios Anastasopoulos, Tonya Thornton, and Hemant Purohit. "PROBER: A System for Real-time Propaganda Behavior Analytics on Social Media and Web Data Streams". In: *2022 IEEE International Conference on Big Data* (*Big Data*). IEEE. 2022, pp. 4812–4815.

[11] **Yasas Senarath**, Jennifer Chan, Hemant Purohit, and Ozlem Uzuner. "Evaluating the Relevance of UMLS Knowledge Base for Public Health Informatics during Disasters". In: *ISCRAM 2021 Conference Proceedings — 18th International Conference on Information Systems for Crisis Response and Management* (*ISCRAM*). 2021.

[12] **Yasas Senarath** and Hemant Purohit. "Evaluating Semantic Feature Representations to Efficiently Detect Hate Intent on Social Media". In: *2020 IEEE 14th International Conference on Semantic Computing* (*ICSC*). IEEE. 2020, pp. 199–202.

[13] Hossein Salemi, **Yasas Senarath**, and Hemant Purohit. "A Comparative Study of Pre-trained Language Models to Filter Informative Code-mixed Data on Social Media during Disasters". In: *ISCRAM 2023 Conference Proceedings — 20th International Conference on Information Systems for Crisis Response and Management* (*ISCRAM*). 2023.

[14] Hossein Salemi, **Yasas Senarath**, Tarin Sultana Sharika, Anuridhi Gupta, and Hemant Purohit. "Summarizing Social Media & News Streams for Crisis-related Events by Integrated Content-Graph Analysis: TREC-2023 CrisisFACTS Track". In: *The Thirty-Second Text REtrieval Conference Proceedings* (*TREC*). 2023, pp. 14–17.

[15] Fedor Vitiugin, **Yasas Senarath**, and Hemant Purohit. "Efficient Detection of Multilingual Hate Speech by Using Interactive Attention Network with Minimal Human Feedback". In: *Proceedings of the 13th ACM Web Science Conference* (*WebSci*). 2021, pp. 130–138.

Human-AI Collaboration

[16] Zinat Ara et al. "Closing the Knowledge Gap in Designing Data Annotation Interfaces for AI-Powered Disaster Management Analytic Systems". In: *Proceedings of the 29th International Conference on Intelligent User Interfaces*. 2024, pp. 405–418.

[17] A Hughes et al. "Human-AI Teaming for COVID-19 Response: A Practice & Research Collaboration Case Study". In: *ISCRAM 2022 Conference Proceedings — 19th International Conference on Information Systems for Crisis Response and Management* (*ISCRAM*). 2022.

[18] Keri Stephens et al. "Online-Computer-Mediated Interviews and Observations: Overcoming Challenges and Establishing Best Practices in a Human-Ai Teaming Context". In: *Proceedings of the 54th Hawaii International Conference on System Sciences* (*HICSS*). 2021.

Opinion Mining

[19] **Yasas Senarath**, Nadheesh Jihan, and Surangika Ranathunga. "A Hybrid Approach for Aspect Extraction from Customer Reviews". In: *The International Journal on Advances in ICT for Emerging Regions* (*ICTer*) 12.1 (2019).

[20] **Yasas Senarath** and Uthayasanker Thayasivam. "DataSEARCH at IEST 2018: Multiple Word Embedding Based Models for Implicit Emotion Classification of Tweets with Deep Learning". In: *Proceedings of the 9th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis* (*WASSA*). Association for Computational Linguistics, 2018, pp. 211–216.

[21] Rajenthiran Jenarthanan, **Yasas Senarath**, and Uthayasanker Thayasivam. "ACTSEA: Annotated Corpus for Tamil & Sinhala Emotion Analysis". In: *2019 Moratuwa Engineering Research Conference* (*MERCon*). IEEE. 2019, pp. 49–53.

[22] Nadheesh Jihan, **Yasas Senarath**, and Surangika Ranathunga. "Aspect Extraction from Customer Reviews Using Convolutional Neural Networks". In: *2018 18th International Conference on Advances in Ict for Emerging Regions* (*ICTer*). IEEE. 2018, pp. 215–220.

[23] Nadheesh Jihan, **Yasas Senarath**, Dulanjaya Tennekoon, Mithila Wickramarathne, and Surangika Ranathunga. "Multi-domain Aspect Extraction using Support Vector Machines". In: *Proceedings of the 29th Conference on Computational Linguistics and Speech Processing* (*ROCLING*). 2017, pp. 308–322.

Community Resilience

[24] **Yasas Senarath**, Steve Peterson, Hemant Purohit, Amanda L Hughes, and Keri K Stephens. "Mining Risk Behaviors from Social Media for Pandemic Crisis Preparedness and Response". In: *Proceedings of the 2021 International Conference on Social Computing, Behavioral-Cultural Modeling, & Prediction and Behavior Representation in Modeling and Simulation* (*SBP-BRiMS*). 2021.

[25] Dante Della Vella et al. "Eliciting Socio-Technical Research Needs on Countering Violence-Inducing and Hateful Speech in Cyberspace". In: *Digital Society* 4.2 (2025), p. 62.

[26] Hoang Long Nguyen, **Yasas Senarath**, Hemant Purohit, and Rajendra Akerkar. "Towards a Design of Resilience Data Repository for Community Resilience". In: *ISCRAM 2021 Conference Proceedings — 18th International Conference on Information Systems for Crisis Response and Management* (*ISCRAM*). 2021, pp. 271–281.

Book Chapters

[27] **Yasas Senarath**, Rahul Pandey, Steve Peterson, and Hemant Purohit. "Citizen-Helper System for Human-Centered AI Use in Disaster Management". In: *International Handbook of Disaster Research*. Springer Nature Singapore Singapore, 2023, pp. 477–497.

Teaching

Database Fundamentals (Teaching Assistant) — Fall 2025, Spring 2026
Conducted hands-on sessions, conducted office hours, evaluated student assignments

Introduction to Research in Applied IT (Teaching Assistant) — Spring 2026
Conducted office hours, evaluated student assignments

Data Analytics in Social Media (Teaching Assistant) — Fall 2025
Developed and conducted hands-on sessions, conducted office hours, evaluated student assignments

Hosted Tutorials

Continual Learning for Online Behavioral Analytics: A Tutorial — Los Angeles, CA, USA
International AAAI Conference on Web and Social Media (ICWSM), Tutorial — 2026

Selected Talks

Human-centered AI Use in Disaster Management using Crowdsourced Data — Castelo Branco, Portugal
4th Workshop on Social Networks and Artificial Intelligence for Disaster Risk Management — 2025

Infrastructure Risk Forecasting using Citizen Sensing & Physical Sensing Data Streams for AI-empowered Disaster Response — Frascati, Italy (Hybrid) 2025
The International Telecommunication Union (ITU)

Early Warning System for Transport Risk Detection — Fairfax, VA, USA
Institute for a Sustainable Earth (ISE) Summer Grad Fellows Professional Development Series — 2025
George Mason University

Detecting Evolving Violence-Inducing Behaviors in the Social-Cyber Space through Continual Learning — Fairfax, VA, USA 2025
Information Sciences and Technology (IST) PhD Symposium
George Mason University

Violence-inducing Behaviour Prevention in Social-Cyber Space — Sogndal, Norway
International Symposium on AI, Data and Digitalization (SAIDD) — 2023

Research Mentorship

Mentored researchers in experimental design, reproducible modeling, research software development, and publication preparation.

Gayatri Ramchandra Vaidya (Ph.D. student) — 2026
Anuridhi Gupta (Ph.D. student) — 2025
Ramya Sreekanta Nayaka (Ph.D. student) — 2024
Varun Velagapuri (master's student) — 2022
Rajenthiran Jenarthanan (undergraduate student) — 2019

Funding Proposal Development

Provost Doctoral Research Scholarship — 2024
Authored the successful research proposal on knowledge-guided continual learning that secured the scholarship from George Mason University

SCC-IRG: Resilient Sheltering Decision Support for Emergency Evacuations using Explainable AI — 2025
Funded by the National Science Foundation (NSF)
Contributed to development and written presentation of the technical approach

Research Systems and Software

CitizenHelper [27]
Designed and developed a human-centered AI system for disaster management.

CrisisFACTS [14]
Developed a system that integrates content and graph analysis to summarize social media and news streams during crisis events.

SinLing
Developed an NLP toolkit for Sinhala language processing (61 GitHub stars).

TweetKit
Created a Python SDK for the Twitter/X API, featured in X's official documentation; collected 2.6M+ tweets used by CitizenHelper and SOCYTI.

TextFlow
Developed a text-annotation platform used in three published disaster-management and COVID-19 studies, including Human–AI evaluations with 28 participants.

Service

Program Committee

AAAI Conference on AI Ethics and Society (AIES)	2024–2026
IEEE Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS)	2024
International Workshop on Cyber Social Threats (CySoc)	2022–2024

Reviewer

Information Processing & Management (IPM)	2026
AAAI Conference on AI Ethics and Society (AIES)	2023
International Conference on Advances in Social Networks Analysis and Mining (ASONAM)	2020–2022
ACM International Conference on Information and Knowledge Management (CIKM)	2022
ACM Conference on Computer Supported Cooperative Work (CSCW)	2024
International AAAI Conference on Web and Social Media (ICWSM)	2021, 2023–2026
ACM Conference on Human-AI Complementarity and Alignment (HCOMP)	2022–2023
ACM Web Conference (WWW)	2021–2022, 2026
ACM Web Science Conference (WebSci)	2020
International Workshop on Cyber Social Threats (CySoc)	2020

Editorial Assistance

AAAI Fall Symposium on AI for Social Good (AI4SG)	2020

Honors and Awards

Provost Doctoral Research Scholarship 2024
Awarded by George Mason University for research in Knowledge-Guided Continual Learning.

Unbounded by Dell Technologies Hackathon 2021
Secured 1st runner-up for CoviSave, a COVID-19 safety application, at the Dell EMC Unbounded Hackathon.

CINTEC Award 2018
Recognized for the highest graded final year project at the Department of Computer Science and Engineering, University of Moratuwa.

Gold Award for Most Outstanding Digital Communication Project 2018
Honored by Rotaract District 3220 Sri Lanka & Maldives for the community initiative TrainLK.